Simple Calculator – Single HTML File

Save this code as **index.html** and open it in a web browser.

```
<!DOCTYPE html>
<html>
<head>
    <title>Simple Calculator</title>
    <style>
        body {
            background: #e0e7ff;
            font-family: Arial;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .calculator {
            background: white;
            padding: 25px;
            border-radius: 15px;
            width: 280px;
        }

        h2 {
            text-align: center;
        }

        input {
            width: 95%;
            height: 45px;
            font-size: 22px;
            margin-bottom: 15px;
            text-align: right;
        }

        .buttons {
            display: grid;
            grid-template-columns: repeat(4, 1fr);
            gap: 8px;
        }

        button {
            padding: 15px;
            font-size: 18px;
            border: none;
            border-radius: 8px;
            background: #dbeafe;
            cursor: pointer;
        }

        button:hover {
            background: #93c5fd;
        }
    </style>
</head>
<body>

    <div class="calculator">
        <h2>Calculator</h2>

        <input type="text" id="display" readonly>

        <div class="buttons">
            <button onclick="clearDisplay()">C</button>
            <button onclick="deleteLast()">DEL</button>
            <button onclick="append('/')">/</button>
            <button onclick="append('*')">*</button>

            <button onclick="append('7')">7</button>
            <button onclick="append('8')">8</button>
            <button onclick="append('9')">9</button>
            <button onclick="append('-')">-</button>

            <button onclick="append('4')">4</button>
            <button onclick="append('5')">5</button>
            <button onclick="append('6')">6</button>
            <button onclick="append('+')">+</button>

            <button onclick="append('1')">1</button>
```

```
            <button onclick="append('2')">2</button>
            <button onclick="append('3')">3</button>
            <button onclick="calculate()">=</button>

            <button onclick="append('0')">0</button>
            <button onclick="append('.')">.</button>
        </div>
    </div>

    <script>
        let display = document.getElementById("display");

        function append(value) {
            if (display.value === "Error") {
                display.value = "";
            }
            display.value += value;
        }

        function clearDisplay() {
            display.value = "";
        }

        function deleteLast() {
            display.value = display.value.slice(0, -1);
        }

        function calculate() {
            try {
                if (!/^[0-9+\\-*/.() ]+$/.test(display.value)) {
                    throw new Error("Invalid input");
                }

                let result = Function(
                    '"use strict"; return (' + display.value + ')'
                )();

                if (!Number.isFinite(result)) {
                    throw new Error("Invalid result");
                }

                display.value = result;
            } catch {
                display.value = "Error";
            }
        }
    </script>

</body>
</html>
```